

APPLICATIONS OF SPECIAL FUNCTIONS IN THE CHARACTERIZATION OF PROBABILITY DISTRIBUTIONS

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1. Summary

In the present paper we consider a general class of probability distributions and study the probability law of the ratio of two independent stochastic variables whose probability laws belong to this general class. As special cases of the ratio distribution we will then obtain the Cauchy law, the Student-t distribution and the F-distribution. Then we will investigate the probability laws of two independent variates if their ratio has, (a) F-distribution, (b) Student-t distribution and (c) Cauchy law. It is shown that there exist an infinite number of independent pairs of stochastic variables whose ratios have the various distributions mentioned above.

2. Introduction

It is well known that the ratio of two independent standard normal variates has a Cauchy distribution. If the ratio of two independent stochastic variables has a Cauchy law, does it necessarily mean that the variates are normal? Several authors have investigated the problem and have given a number of counter examples, see Fox (1965), Laha (1958), Laha (1959), Mouldon (1956) and Steck (1958). Further, the ratio of two independent chi-square variables has the F-distribution. Does it necessarily follow that the variables are chi-squares if the ratio has an F-distribution? It is also evident that the ratio of a normal variate to the square root of an independent chi-square variate, has a Student-t distribution. Does it mean that if the ratio of two independent stochastic variables has a Student-t distribution then one is to be a normal and the other is a chi-square variate? We will investigate all these problems with reference to a general class of choices for all the cases mentioned above.

The standard normal law, the student-t law and the F-distribution will be taken in the following forms for convenience.

Normal: $f_1(x) = (2/\pi)^{1/2} \cdot \exp(-x^2/2)$, $0 < x < \infty$

Student-t: $f_2(t) = 2 \Gamma(\beta/2 + 1/2) / (\pi)^{1/2} \Gamma(\beta/2) (1 + t^2)^{-(\beta+1)/2}$, $0 < t < \infty$

F-distribution: $f_3(x) = \Gamma(\beta_1 + \beta_2) x^{\beta_1-1} / \Gamma(\beta_1) \Gamma(\beta_2) (1+x)^{-(\beta_1+\beta_2)}$, $0 < x < \infty$

where $f_1(x) = 0 = f_2(x) = f_3(x)$ for $x \leq 0$; $\beta > 0$, $\beta_1 > 0$, and $\beta_2 > 0$.

In what follows the following notations will be used. The following is the H-function, which is even more general than the Meijer G-function (Erdelyi, 1953, pp. 207), introduced by Fox (1961) in connection with the investigation of

symmetric Fourier kernels of H-functions. The H-function is defined by means of the Mellin-Barnes integral,

$$(1) \quad H_{p,q}^{m,n} \left[x \left| \begin{matrix} (a_p, A_p) \\ (b_q, B_q) \end{matrix} \right. \right] = (1/2 \pi i) \int_L \frac{\prod_{k=1}^m \Gamma(b_k + B_k s) \prod_{k=1}^n \Gamma(1 - a_k - A_k s)}{\prod_{k=m+1}^q \Gamma(1 - b_k - B_k s) \prod_{k=n+1}^p \Gamma(a_k + A_k s)} x^{-s} ds.$$

where an empty product is interpreted as unity; $0 \leq m \leq q$, $0 \leq n \leq p$; A_k 's and B_k 's are positive; all the poles of the integrand are simple; the contour extends from $c - i\infty$ to $c + i\infty$ such that all the poles of $\Gamma(b_k + B_k s)$, $k = 1, \dots, m$ lie on the right of the contour and those of $\Gamma(1 - a_k - A_k s)$, $k = 1, \dots, n$ lie on the left of it. Fox (1961) has shown that,

$$(2) \quad \int_0^\infty x^{s-1} H_{p,q}^{m,n} \left[a x \left| \begin{matrix} (a_p, A_p) \\ (b_q, B_q) \end{matrix} \right. \right] dx = \frac{X(s)}{a^s}$$

where $X(s)$ is the coefficient of x^{-s} in the integrand of (1), $a > 0$, $R(b_k + B_k s) > 0$, for $k = 1, \dots, m$ and $R(1 - a_k - A_k s) > 0$ for $k = 1, \dots, n$ and $R(\cdot)$ denotes the real part of $(\cdot)$. Define

$$(3) \quad \Phi(\sigma_j) = \frac{1}{\chi\left(\frac{\sigma_j}{\epsilon}\right)}, \quad j = 1, 2 \text{ for } \epsilon, \sigma_j > 0.$$

3. A general class of probability functions

Here we will consider a general class of probability functions defined by the H-functions and we will study the distributions of the ratio of independent stochastic variables whose probability laws belong to this general class.

Theorem 1. Let X_1, X_2 be two independent stochastic variables having the probability laws

$$(4) \quad P_j(x) = r \theta^{\sigma_j/r} \theta(\sigma_j) x^{\sigma_j-1} H_{p,q}^{m,n} \left[\theta x^r \left| \begin{matrix} (a_p, A_p) \\ (b_q, B_q) \end{matrix} \right. \right] \quad \text{for } x > 0$$

$= 0 \text{ elsewhere.}$

where, $j = 1, 2$; $\theta > 0$, $0 < r < \infty$, $\sigma_1, \sigma_2 > 0$, and the other parameters have the same meanings given in the preceeding section, then $W = X_1/X_2$ has the probability function,

$$(5) \quad P_W(x) = r x^{-\sigma_2-1} \phi(\sigma_1) \phi(\sigma_2) H_{p+q, p+q}^{m+n, m+n} \left[x^r \left| \begin{matrix} (1-b_q, B_q), (w_p, A_p) \\ (z_q, B_q), (1-a_p, A_p) \end{matrix} \right. \right]$$

for $x > 0$ and $P_W(x) = 0$ elsewhere, where,

$$r w_n = r a_n + A_n(\sigma_1 + \sigma_2) \text{ and } r z_m = r b_m + B_m(\sigma_1 + \sigma_2).$$

Proof: We have $W = X_1/X_2$. Let $V = \log W = \log X_1 - \log X_2$.

The characteristic function of $\log X_j$ is,

$$(6) \quad f_{X_j}(t) = \epsilon \theta^{\sigma_j/\epsilon} \phi(\sigma_j) \int_0^\infty x^{it+\sigma_j-1} H_{p,q}^{m,n} \left[\theta x^\epsilon \left| \begin{matrix} (a_p, A_p) \\ (b_q, B_q) \end{matrix} \right. \right] dx$$

$$= \frac{\phi(\sigma_j)}{\theta \frac{it}{\epsilon} \theta(\sigma_j + it)}$$

by virtue of (2). The characteristic function of V is, therefore,

$$(7) \quad f_V(t) = \theta(\sigma_1) \theta(\sigma_2) / \theta(\sigma_1 + it) \theta(\sigma_2 - it).$$

The probability function for V is,

$$(8) \quad P_V(x) = (1/2) \theta(\sigma_2) \phi(\sigma_1) \int_{-\infty}^\infty e^{-itx} dt / \theta(\sigma_1 + it) \theta(\sigma_2 - it)$$

$$= (1/2) \theta(\sigma_1) \phi(\sigma_2) e^{-\sigma_2 x} \int_{-\sigma_2 - i\infty}^{-\sigma_2 + i\infty} e^{-zx} dz / \theta(\sigma_1 + \sigma_2 + z) \phi(-z)$$

by putting $it - \sigma_2 = z$. Now replacing x by $\log x$, we see that,

$$(9) \quad P_W(x) = x^{-\sigma_2-1} \phi(\sigma_1) \phi(\sigma_2) \int_{-\sigma_2 - i\infty}^{-\sigma_2 + i\infty} x^{-z} dz / \theta(\sigma_1 + \sigma_2 + z) \phi(-z).$$

On interpreting the integral on the right by (1) we get (5).

Corollary 1. When $A_p = B_q = 1$ for all p and q , the H -function reduces to the Meijer G -function (Erdelyi, 1953, p. 207) and the theorem gives rise to the following interesting and generalized result. If two independent stochastic variables X_1 and X_2 have the probability laws given by,

$$P_j(x) = r \theta^{\sigma_j/r} \phi_1(\sigma_j) x^{\sigma_j-1} G_{p,q}^{m,n} \left[\theta x^r \left| \begin{matrix} (a_p) \\ (b_q) \end{matrix} \right. \right] \quad \text{for } x > 0, j = 1, 2;$$

$$= 0 \text{ elsewhere,}$$

where $\phi_i(\sigma_j) = \phi(\sigma_j)$ with $A_j = B_i = 1, i = 1, \dots, q; j = 1, \dots, p$, then X_1/X_2 has the probability law,

$$P_W(x) = r x^{-\sigma_2-1} \phi_1(\sigma_1) \phi_1(\sigma_2) G_{p+q, p+q}^{m+n, m+n} \left[x^r \left| \begin{matrix} (1-b_q), (w'_p) \\ (z'_q), (1-a_p) \end{matrix} \right. \right] \text{ for } x > 0,$$

$$= 0 \text{ elsewhere,}$$

where, $w'_n = r a_n + (\sigma_1 + \sigma_2)$ and $z'_m = r b_m + (\sigma_1 + \sigma_2)$.

Corollary 2. When X_1 and X_2 are independent and have the standard normal distribution then W has the Cauchy law. This follows from (9) by taking the special case, $m = 1$, $n = 0$, $p = 0$, $q = 1$, $r = 2$, $\sigma_1 = \sigma_2 = 1$, $\theta = 1/2$, $b_1 = b_2 = 0$. Since the H-function then yields $\exp(-x^2/2)$, $P_j(x)$ reduces to the normal law for $j = 1, 2$.

Corollary 3. If X_1 and X_2 are independent and have the gamma distribution with the parameters σ_1 and σ_2 then $W = X_1/X_2$ has the law,

$$P_W(x) = \Gamma(\sigma_1 + \sigma_2) x^{\sigma_1-1} (1+x)^{-\sigma_1-\sigma_2} / \Gamma(\sigma_1) \Gamma(\sigma_2) \text{ for } x > 0,$$

= 0 elsewhere.

This can be deduced from (9) by taking $m = 1$, $n = 0$, $p = 0$, $q = 1$, $r = 1$, $A_p = B_q = 1$ for all p and q , $b_j = \sigma_j - 1$. In this case the H-function reduces to the exponential form. The F-distribution is also obtained under these conditions by taking appropriate values of σ_1 and σ_2 .

Corollary 4. The student-t distribution as given in section 2 can be obtained by taking $\theta = 1/2$, $b_j = \sigma_j - 1$, $r = 2$, $\sigma_1 = 1$, $\sigma_2 = \sigma^2$, $m = 1$, $n = 0$, $p = 0$, $q = 1$. Here the probability function $P_1(x)$ of (4) reduces to the normal form and $P_2(x)$ reduces to a modified gamma form.

Corollary 5. When $r = 2$, $\theta = 1/2$, $\sigma_1 = \sigma_2 = 1$, $p = 1$, $q = 2$, $m = 1$, $n = 1$, $a_1 = 0$, $b_1 = v/2$, $b_2 = -v/2$ and using the formula (Erdelyi, 1954b, p. 435),

$$\Gamma(a) x^b {}_1F_1(a; c; -x) / \Gamma(c) = G_{1,2}^{1,1} \left[x \mid \begin{matrix} 1-a+b \\ b, 1-c+b \end{matrix} \right]$$

(4) reduces to the function $N_v(x)$ defined by Fox (1965). The ratio of any two independent stochastic variables having density functions in the class of $N_v(x)$, has the Cauchy law, where,

$$(10) \quad N_v(x) = x^v e^{-x^2/2} \Gamma(v/2 + 1) {}_1F_1(v/2; v + 1; x^2/2) / (2\pi)^{1/2} v^{1/2} \Gamma(v+1)$$

for $x > 0$ and is zero elsewhere.

4. A family of probability distributions having the same ratio properties

We know that the ratio of two independent gamma variates has a F-distribution as given in section 2. In theorem 2 we will prove that this ratio property is not only enjoyed by the gamma variates but also by a general class of probability functions

$$(11) \quad \text{Let } M_{a_i, v}(x) = \frac{x^v \Gamma(a_i + 1 + v)}{2^v \Gamma(v+1) \Gamma(a_i)} {}_2F_1((a_i + v + 1)/2;$$

$$(a_i + v + 2)/2; v + 1; -x^2) \text{ for } x > 0$$

$$= 0 \text{ elsewhere for } j = 1, 2.$$

Lemma 1. $\int_0^\infty M_{a_i, v}(x) dx = 1$ for $i = 1, 2$.

The proof follows from the following result, (Erdelyi, 1954a, p. 336),

$$\int x^{s-1} {}_2F_1(\alpha; \beta; x) dx = B(s, \alpha-s) B(s, \beta-s) / B(s, \beta-s)$$

where $R(s) > 0$, $R(\alpha-s) > 0$, $R(\beta-s) > 0$, B denotes the well-known Beta function, ${}_2F_1$ denotes the ordinary hypergeometric function (Erdelyi, 1953, p. 56) and $0 < R(s) < \min \{R(\alpha), R(\beta)\}$.

The following is the Parseval property of Hankel transform (Sneddon, 1951, theorem 23, p. 60), which will be used in our investigations that follow.

If $\int_0^\infty t J_v(pt) \phi(t) dt = \psi_1(p)$ and

$$\int_0^\infty t J_v(pt) \rho(t) dt = \psi_2(p) \text{ then,}$$

$$(12) \int_0^\infty t \phi(t) \rho(t) dt = \int_0^\infty p \psi_1(p) \psi_2(p) dp$$

where $J_v(x)$ is the ordinary Bessel function (Erdelyi, 1954b, p. 426) and $\phi(t)$ and $\rho(t)$ are functions such that the integrals exist.

Theorem 2. If two independent stochastic variables X_1 and X_2 have the probability function $M_{a_i, v}(x)$ for $i = 1, 2$ then X_1/X_2 has the F-distribution as given in section 2, for all values of v such that $M_{a_i, v}(x)$ exists.

Proof: The theorem can be proved if we can establish the following relations.

$$(13) \int_0^\infty M_{a_1, v}(x) dx = \int_0^\infty M_{a_2, v}(x) dx = 1,$$

$$(14) \int_0^\infty u M_{a_1, v}(ux) M_{a_2, v}(u) du = \Gamma(a_1 + a_2) x^{a_2-1} / \Gamma(a_1) \Gamma(a_2) (1+x)^{-(a_1+a_2)}$$

for all v . (Since X_1 and X_2 are independent the probability law of X_1/X_2 should satisfy the left side of (14)).

Proof: Lemma 1 proves (13) and we know from (Erdelyi, 1954b, p. 29) that,

$$(15) \int_0^\infty t J_v(ut) (t)^{a_i-1} e^{-t} dt / \Gamma(a_i) = M_{a_i, v}(u)$$

where $R(a_i + v) > -1$, $R(u) > 0$, $p > 0$, and,

$$(16) \int_0^\infty t J_v(ut) (t)^{a_i-1} e^{-t/a} dt / \Gamma(a_i) = M_{a_i, v}(au). a^{a_i+1}$$

Now applying (12) we obtain,

$$\begin{aligned} \int_0^\infty u M_{a_1, v}(au) M_{a_2, v}(u) du &= \int_0^\infty \frac{t t^{a_1-1} e^{-t/a}}{\Gamma(a_1) a^{a_1+1}} \cdot \frac{t^{a_2-1} e^{-t}}{\Gamma(a_2)} dt \\ &= \frac{\Gamma(a_1+a_2)}{\Gamma(a_1)\Gamma(a_2)} \cdot \frac{a^{a_2-1}}{(1+a)^{a_1+a_2}} \text{ for all } v \end{aligned}$$

Hence the result.

Lemma 2. $\int_0^\infty N_{a_i, v}(x) dx = 1$

where,
$$N_{a_i, v}(x) = \frac{\Gamma\{(a_i+v+1)/2\} x^v {}_1F_1\{(a_i+v+1)/2; v+1; -x^2/4\}}{2^v \Gamma(v+1) \Gamma(a_i/2)}$$

for $x > 0$

Proof: By putting $x = 2u^{1/2}$ we obtain

$$\begin{aligned} (18) \quad \int_0^\infty N_{a_i, v}(x) dx &= \frac{\Gamma\{(a_i+v+1)/2\}}{\Gamma(v+1)\Gamma(a_i/2)} \int_0^\infty u^{(v+1)/2-1} \\ &\quad \cdot {}_1F_1\{(a_i+v+1)/2; v+1; -u\} du \end{aligned}$$

Since we know that (Erdelyi, 1954b, p. 435),

$$(19) \quad G_{1,2}^{1,1}\left[t \mid \begin{matrix} 1-a \\ 0, 1-b \end{matrix}\right] = \Gamma(a) {}_1F_1(a; b; t)/\Gamma(b)$$

(18) transforms into

$$\begin{aligned} (20) \quad \frac{\Gamma\{(a_i+v+1)/2\}\Gamma(v+1)}{\Gamma(v+1)\Gamma\{(a_i+v+1)/2\}} &\Gamma(a_i/2) \int_0^\infty G_{1,2}^{1,1}\left[t \mid \begin{matrix} 1-a_i-v \\ 0, -v \end{matrix}\right] \\ &t^{(v+1)/2-1} dt \end{aligned}$$

Using a result (Erdelyi, 1954a, p. 337), we obtain

$$(21) \quad \int_0^\infty G_{1,2}^{1,1}\left[t \mid \begin{matrix} 1-a_i-v \\ 0, -v \end{matrix}\right] t^{(v+1)/2-1} dt = \Gamma(a_i/2)$$

We establish the required result from (20) and (21).

Theorem 3. If X_1 and X_2 are two independent stochastic variables having the probability functions $N_{a_i, v}(x)$ for $i = 1, 2$ then X_1/X_2 has the probability law,

(22) $g(a) = 2\Gamma\{(a_1+a_2)/2\} a^{a_2-1} / \Gamma(a_1/2) / (1+a^2)^{a_1+a_2/2}$
for $a > 0$ and for all v .

Proof: From (Erdelyi, 1954b, p. 30) it follows that

$$(23) \int_0^\infty \left[x \left| \frac{2x^{a_1-1} e^{-x^2/a^2}}{a^{a_1+1} \Gamma(a_1/2)} \right| \right] J_v(ux) dx = N_{a_1, v}(ua)$$

where $R(a_1) > 0$, $R(a^2) > 0$, $u > 0$ and,

$$(24) \int_0^\infty x \left[2x^{a_2-1} e^{-x^2} / \Gamma(a_2/2) \right] J_v(ux) dx = N_{a_2, v}(u).$$

From (12) we find that,

$$(25) \int_0^\infty u N_{a_1, v}(ua) N_{a_2, v}(u) du = 4 \int_0^\infty \frac{e^{-(1+1/a^2)x^2} x^{a_1+a_2-1} dx}{\Gamma(a/2) \Gamma(a_2/2) a^{a_1+1}} \\ = 2\Gamma\{(a_1+a_2)/2\} a^{a_2-1} / \Gamma(a_1/2) \Gamma(a_2/2) (1+a^2)^{(a_1+a_2)/2} \text{ for all } v.$$

Corollary 1. When $a_1 = \sigma$ and $a_2 = 1$ the right side of (25) reduces to the Student-t distribution given in section 2. Here we get an infinite number of pairs of independent stochastic variables for which the ratio has a Student-t distribution written in a suitable form.

Corollary 2. When $a_1 = a_2 = 1$ the right hand side of (25) reduces to the following law given by,

$$f(x) = 2/\pi (1+x^2) \text{ for } x > 0 \text{ and } f(x) = 0 \text{ elsewhere.}$$

This is the Cauchy law written in an appropriate form. It is easy to notice that Theorem 3 can be derived from Theorem 2 but Theorem 3 gives a family of distributions in terms of ${}_1F_1(a, \beta, x)$ and hence an independent derivation is given here because of the importance of ${}_1F_1(a, \beta, x)$ in statistical distributions.

Remark 1. From the nature of $P_j(x)$ of (4) and $P_W(x)$ of (5) it is evident that both belong to the same family of functions.

Remark 2. Even if the parameters are all different in $P_1(x)$ and $P_2(x)$, still the ratio distribution will have the same structure as in $P_W(x)$ with corresponding changes in the parameters. This has already been used in corollaries 3 and 4 of Theorem 1.

Remark 3. Since $P_j(x)$ is defined in terms of H-functions it is apparent that the ratio distribution, of any two independent stochastic variables whose density functions can be expressed in terms of any single special function, is included in $P_W(x)$ as a special case.

For example, the functions involved in defining the densities in Theorems 2 and 3 are special cases of H-functions. Hence, structurally Theorems 2 and 3 are contained in Theorem 1. But Theorems 2 and 3 serve a different purpose than that of Theorem 1. That is, Theorems 2 and 3 are used to show the non-uniqueness of some ratio distributions whereas Theorem 1 gives a general ratio distribution.

Remark 4. The results are valid for those choices of the parameters which yield non-negative probability density functions.

REFERENCES

- ERDELYI, A. et. al. (1953). *Higher Transcendental Functions*, Vol. I. McGraw-Hill, New York.
- ERDELYI, A. et. al. (1954a). *Tables of Integral Transforms*. Vol. I, McGraw-Hill, New York.
- ERDELYI, A. et. al. (1954b). *Tables of Integral Transforms*. Vol. II, McGraw-Hill, New York.
- FOX, C. (1961). The G and H functions as symmetrical Fourier kernels. *Trans. Amer. Math. Soc.* 98, 395-429.
- FOX, C. (1965). A family of distributions with the same ratio property as normal distribution. *Canad. Math. Bull.* 8, 631-635.
- LAHA, R.G. (1958). An example of a non-normal distribution where the quotient follows the Cauchy law. *The National Academy of Sciences (U.S.A.)* 44, 222-223.
- LAHA, R.G. (1959). On the laws of Cauchy and Gauss. *Ann. Math. Statist.* 30, 1165-1174.
- LUKACS, E. and LAHA, R.G. (1964). *Applications of Characteristic Functions*. Charles Griffin and Company, London.
- MOULDON, J.G. (1956). Characterizing properties of statistical distributions. *Quarterly J. Math. (Oxford) Series 2*, 7, 155-160.
- SNEDDON, I.N. (1951). *Fourier Transforms*. McGraw-Hill, New York.
- STECK, G.P. (1958). An uniqueness property not enjoyed by the Normal distribution. *Ann. Math. Statist.* 29, 604-606.